

Particulate Matter Monitoring & Health Effects

Issue 2026-08-06 | Entry window 6 August 2026 | 27 records in scope

27RECORDS IN SCOPE
(23 rejected, 9 held)**7**SENSING RECORDS
(26% OF CORPUS)**8**SUBTOPIC CLUSTERS
REPRESENTED**9**TIER-A
RECORDS**6**EFFECT ESTIMATES
EXTRACTED

Signal of the day

What people *smell* moves protective behaviour roughly five times more than what the network *measures*. The last metre of a monitoring system is perceptual, and it is currently unengineered.

Slavik et al. (*Environ Challenges*) linked survey responses from **2,086 parents** in wildfire-prone western North America to three and a half years of smoke-monitoring data (2020–2023). Parents who *perceived* smoke exposure took **1.11 more protective actions** (95% CI 0.92–1.30) than those who did not, whereas each **10 $\mu\text{g}/\text{m}^3$ of measured $\text{PM}_{2.5}$** on smoke days bought only **0.23 additional actions** (0.06–0.40) — a ~ 4.8 -fold gap in favour of the subjective channel. The measured-exposure response is also non-monotonic: actions rise, plateau at moderate concentrations, then rise again. Since sub-visible smoke is demonstrably harmful, a network that reports accurate concentrations into a population that acts on sight and smell is delivering a fraction of its achievable health benefit. Read against the two records in today's corpus that show *mass* is the wrong currency for the hazard — Yadav et al.'s Agra year-long series, where PM_{10} mass and endotoxin are effectively uncorrelated ($r_s = 0.10$, $p = 0.75$) across an 8-fold seasonal endotoxin swing, and Lei et al.'s eight-year speciated PM_1 record in Dublin — the day's through-line is that the two ends of the measurement chain are both weakly coupled to the thing that matters. Composition is not mass, and communicated risk is not measured risk.

What changed today

- **The confidence-interval drought broke.** Jung et al. (*Ann Rheum Dis*) report the first peer-reviewed PM ratio estimate with an interval this watch has seen in five issues: **OR 1.113 (95% CI 1.017–1.218)** for rheumatoid arthritis flare per monthly mean $\text{PM}_{2.5}$, across 12,583 outpatient visits from 1,070 patients, with a within-patient case-crossover sensitivity arm that reproduces the association at >2 weeks of cumulative exposure. $\text{PM}_{2.5}$ was the only one of six pollutants to survive. Six effect estimates carry intervals today against one on 5 August.
- **Two well-powered nulls, and both are informative rather than empty.** Sewor et al. found *no* association between 48-hour personal $\text{PM}_{2.5}$ and blood pressure in 622 rural Rwandan children — at a **median exposure of 192.9 $\mu\text{g}/\text{m}^3$** , roughly $40\times$ the WHO annual guideline. The authors' own reading is the right one: the exposure–response may simply be flat that far up the global continuum, which is a statement about where monitoring effort buys epidemiological information. Separately, Lees et al. (preprint) report a $\text{PM}_{2.5}$ -component mixture IRR of **1.31 (0.99–1.73, $p = 0.06$)** for hospitalisation in 2,511 CKD patients with **93% of the WQS weight on nitrate alone** — a null whose weight vector is more interesting than its point estimate.
- **Instrumentation side: three papers about the limits of what an instrument can be asked to mean.** Liu et al. (*ACP*) show that inferring the aerosol mixing-state index χ from HTDMA κ distributions works until externally mixed *intermediate*-hygroscopicity species violate the method's binary assumption, and publish size-resolved error bounds so the retrieval can be used honestly. Tian et al. (*AMT*) do the analogue for lidar-retrieved CCN profiles with a heterogeneity index that flags when vertical layering has invalidated the homogeneous-composition assumption. Yee et al. (*AMT*) release UCB-GLOBES, $\sim 27,000$ GC/EC-MS spectra — and report **$\sim 70\%$ of them still match nothing**, in their own library or in NIST.
- **Imputation gets an honest negative.** Somnugpong's similarity-weighted seasonal-anchor Stineman method improves on plain Stineman for the UCI dataset under MCAR and MAR but *does not* beat it on Beijing $\text{PM}_{2.5}$. The paper is framed as a boundary condition rather than a method launch, which is the correct framing and a rare one — relevant to anyone reconstructing gappy low-cost sensor series.
- **Preclinical cardiac remodelling is now measurable at IQR-scale exposure contrasts.** Leiser et al. (*JACC Adv*) find per-IQR $\text{PM}_{2.5}$ associated with $+0.52\%$ global longitudinal strain (0.20–0.83), -1.05% LVEF per year (-1.73 to -0.37) and -1.39 left atrial volume index (-2.25 to -0.54) over a median 4.3 years of serial echocardiography in 1,630 people. Note the sign pattern is internally awkward — GLS and LAVI move in directions that do not obviously cohere — and the paper does not resolve it.
- **Sub-Saharan Africa contributed three records, the most in any window so far** — Mushore et al. (road-edge gradient *reverses sign* between sites), Makanjuola (563 street sweepers) and Sewor above, all three using handheld or wearable optical instruments as the primary exposure measurement.

Corpus analytics

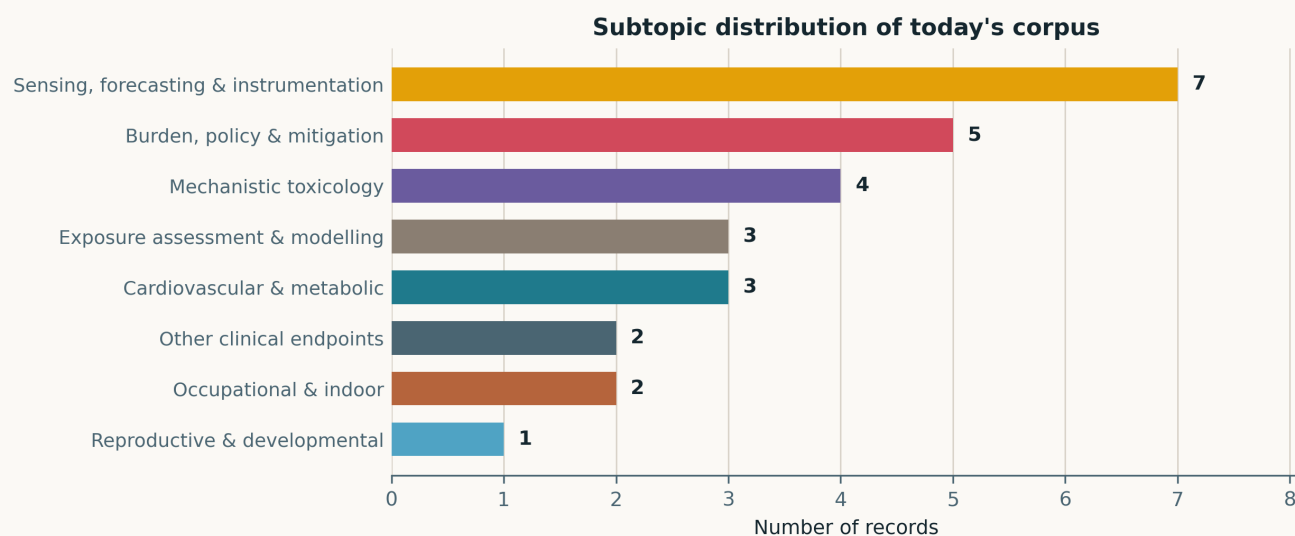


Fig. Subtopic assignment of the 27 in-scope records, single-label by dominant contribution. *Sensing, forecasting & instrumentation* leads with 7 (26%), followed by *burden/policy* (5) and *mechanistic toxicology* (4). Eight of the ten standing clusters are populated — the widest coverage since 27 July — but *Neuro / mental health* and *Respiratory & allergic* are both empty as dominant labels, which is an artefact of assignment rather than of the corpus: four records carry a respiratory endpoint but sit under sensing or occupational because that is where their methodological contribution lies.

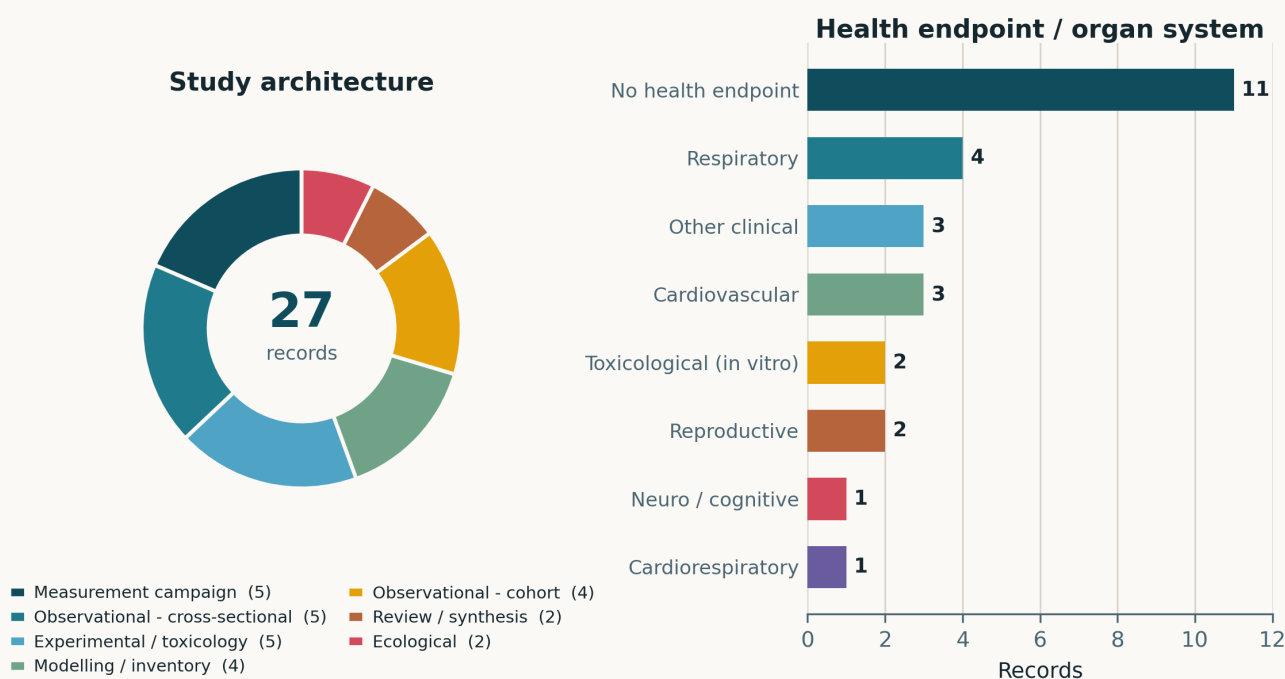


Fig. Left — study architecture, evenly spread for the first time in a fortnight: measurement campaigns, cross-sectional observational work and experimental toxicology tie at 5 each, with modelling/inventory and cohort designs at 4. Right — health endpoint by organ system. **Eleven of 27 records carry no human health endpoint**, consistent with an instrumentation-heavy window; respiratory (4) leads the endpoints that exist, ahead of cardiovascular (3) and other-clinical (3). There is no trial or intervention record in this window at all.

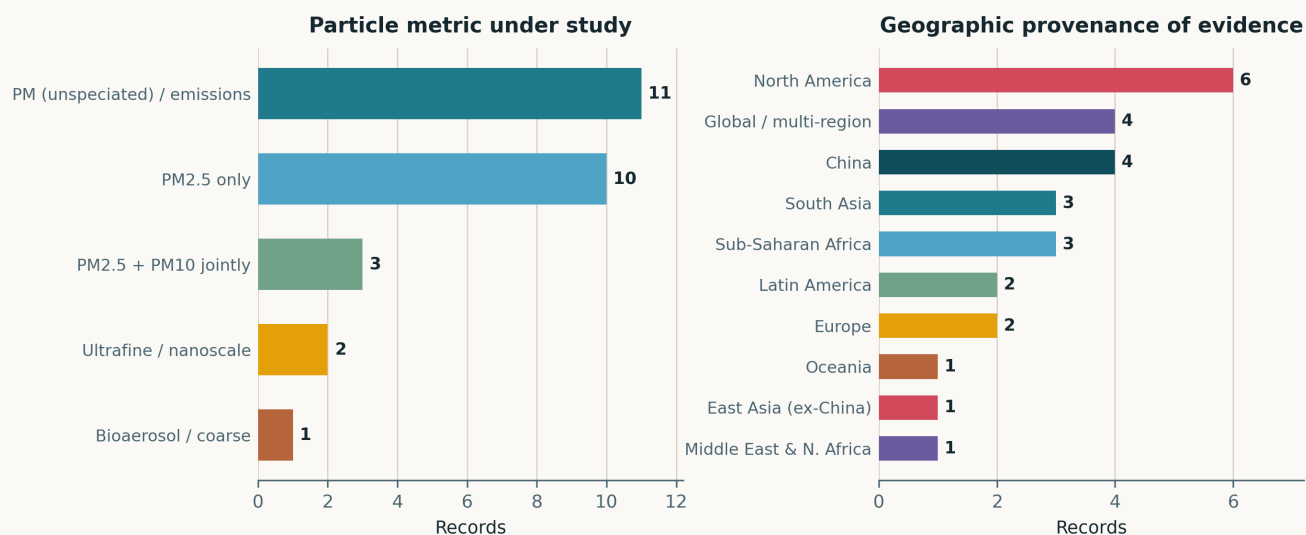


Fig. Left — particle metric. Ten records use PM_{2.5} alone and three pair it with PM₁₀. The 11-record unspeciated/emissions bar is genuine heterogeneity rather than missing detail — it holds PM₁, CCN number, mass-spectral organic aerosol, a mixing-state index and two records whose exposure is a gas. Two records are resolved to ultrafine/nanoscale and one to coarse bioaerosol, the latter being the Agra endotoxin series. Right — geographic provenance. North America leads (6), then China and the global/multi-region group (4 each). **Sub-Saharan Africa contributes 3 and South Asia 3** — together 22% of the corpus, which is the strongest representation from the highest-burden regions this watch has recorded.

Life-course windows addressed today (records may span several stages)

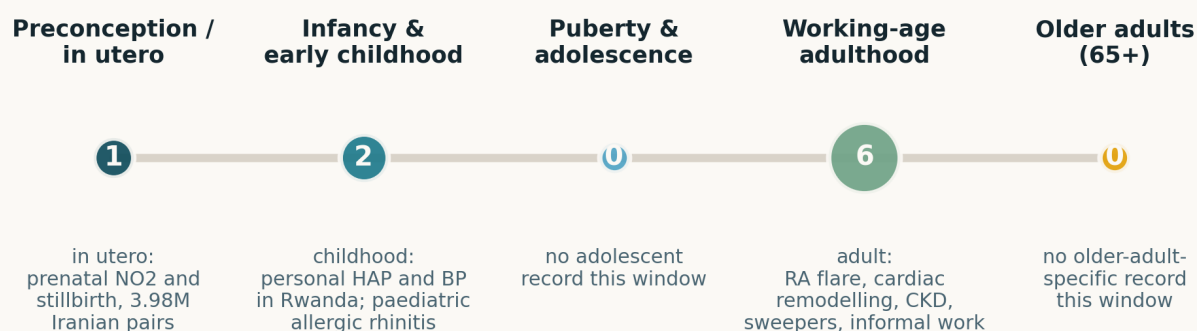


Fig. Life-course windows addressed; counts are non-exclusive. Nine of 27 records carry a human life-course window. Adults dominate (6). **The adolescent and older-adult windows are both empty**, the older-adult gap now running three consecutive issues despite that group carrying the largest cardiovascular attributable burden. The single in-utero record is a nitrogen-dioxide study, not a PM study.

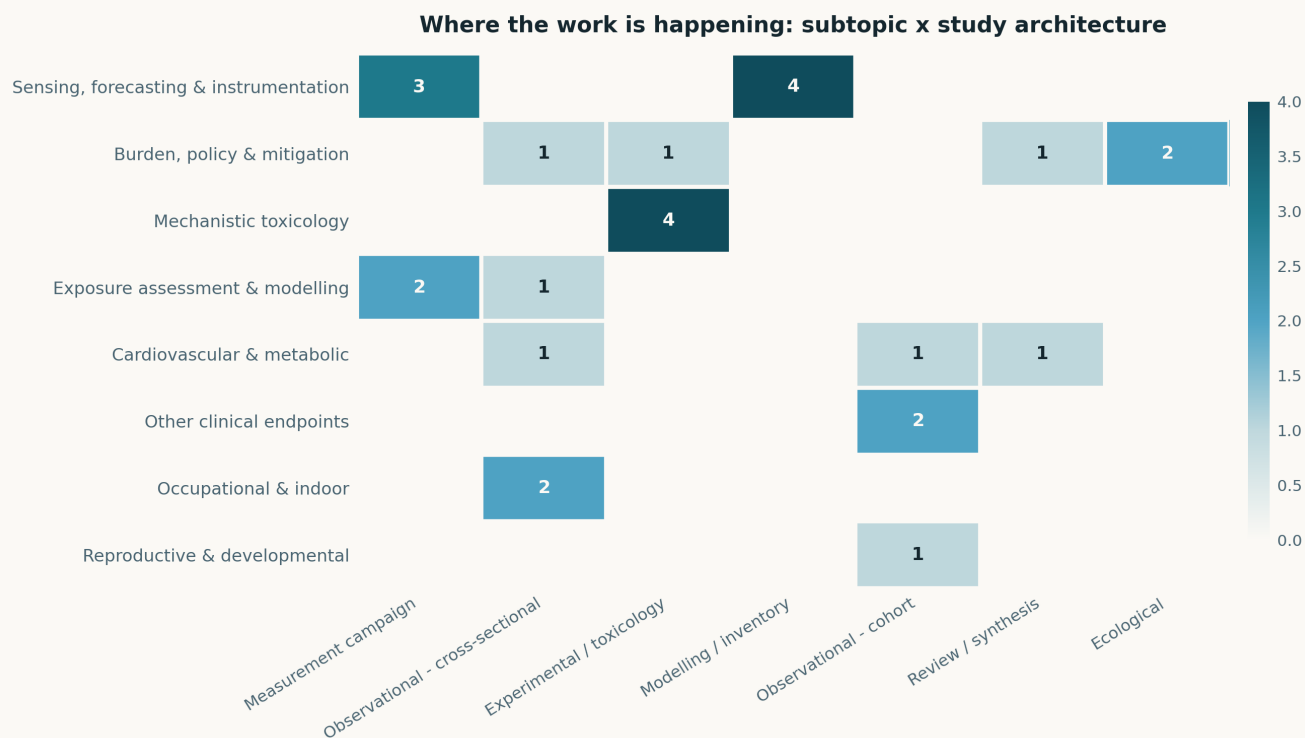


Fig. Subtopic × study-architecture cross-tabulation. The structural feature to note is that the *Sensing* row is split between measurement campaigns and modelling with **no cell in any observational-epidemiology column** — instrumentation and health inference remain disjoint within this window, as they have been in every issue since 27 July. Mechanistic toxicology is again wholly experimental (4/4), with no human biomarker bridge.

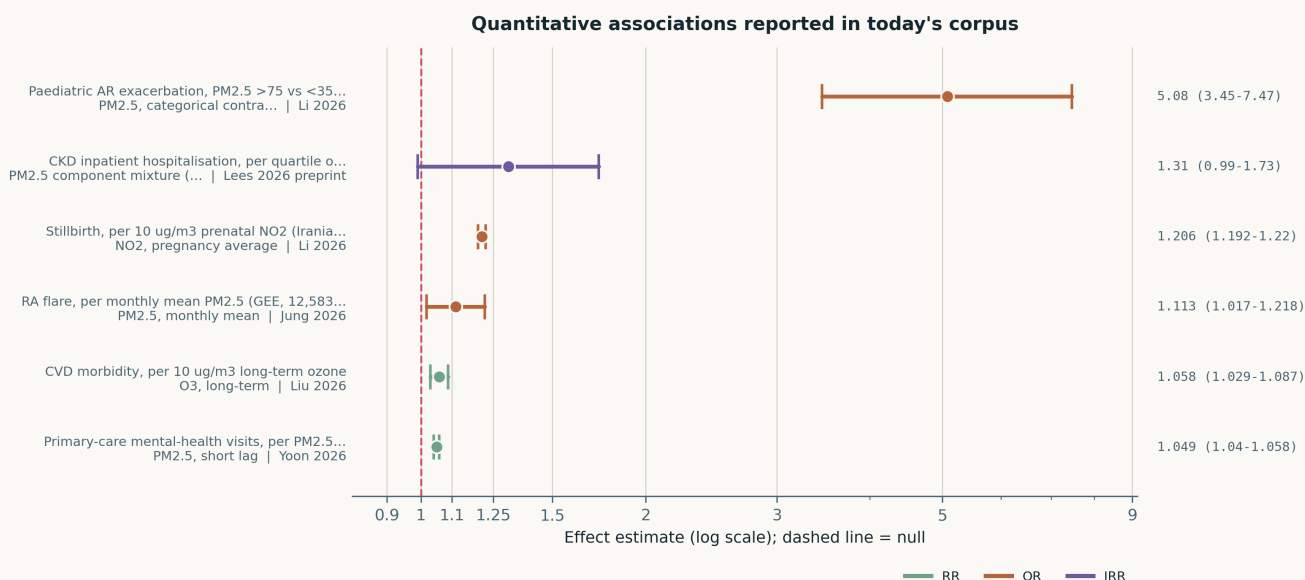


Fig. All six effect estimates carrying an interval, on a common logarithmic axis. **Exposure contrasts are heterogeneous and the estimates are not mutually comparable:** Jung is per monthly mean PM_{2.5}, Li (*Front Pediatr*) is a categorical >75 vs <35 $\mu\text{g}/\text{m}^3$ contrast, Yoon is a short-lag RR per unit bushfire-smoke PM_{2.5}, Lees is per quartile of a weighted-quantile-sum component index, and the two lowest entries are **co-pollutant estimates** (NO₂, O₃) and not PM at all — they are plotted so a rollup does not read their absence as pipeline loss. Read direction and precision, not relative magnitude. Four of the six are PM estimates, ending a four-issue run in which no peer-reviewed PM ratio estimate with a confidence interval entered the corpus.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase.

Sensing, forecasting & instrumentation

7 records

Somnugpong 2026 Environ Monit Assess

Similarity-weighted seasonal-anchor Stineman interpolation for gappy air-quality series

SW-SAST keeps a local Stineman anchor and selectively borrows cross-day information when nearby daily analogues are compatible with the local estimate. Evaluated on UCI Air Quality and Beijing PM_{2.5} under MCAR, MAR and MNAR at 10–50% missingness, 30 repeated runs per condition, against a univariate LSTM baseline and with contiguous gap-length experiments. **It improves on plain Stineman for UCI, chiefly under MCAR and MAR, and fails to beat it on Beijing.** The contribution is explicitly a boundary condition on when seasonal borrowing helps, not a replacement interpolator — which is the right claim and an unusually disciplined one. Directly applicable to reconstructing low-cost-sensor series where dropouts are neither random nor uniform.

doi: 10.1007/s10661-026-15770-8

Lei et al. 2026 Atmos Chem Phys

Eight-year speciated PM₁ record in Dublin: heating emissions versus transboundary transport

Continuous chemically speciated PM₁, 2016–2023, resolving two distinct pollution regimes: short intense events (hours, PM₁ >100 µg/m³) driven by local solid-fuel home heating, and moderate multi-day events (<60 µg/m³) from continental transport. Annual PM₁ fell from 6.5 to <5.0 µg/m³, led by nitrate and ammonium (−0.11 and −0.09 µg m^{−3} yr^{−1}). **Transported oxygenated organic aerosol rose (+0.34 µg m^{−3} yr^{−1})** while locally formed OOA barely fell, so the composition shifted even as mass declined — and 2023 showed a heating-driven rebound. A demonstration that a mass-trend headline can be exactly wrong about what a network is now measuring.

doi: 10.5194/acp-26-10927-2026

Liu et al. 2026 Atmos Chem Phys

From κ to χ : particle-resolved evaluation of hygroscopicity-based mixing-state retrieval

PartMC-MOSAIC urban-plume ensembles spanning emissions, ageing and meteorology were used to compare the particle-resolved reference mixing-state index against the Yuan–Zhao retrieval from HTDMA κ distributions, per diameter over 50–250 nm. The retrieval performs well for many populations but **systematically overestimates χ when externally mixed intermediate-hygroscopicity components break its binary more-/less-hygroscopic assumption.** The useful output is the size-resolved error distribution, which the authors then apply as uncertainty bounds to long-term HTDMA records at urban, continental and coastal sites. Model-derived error bars on an observational retrieval — the validation is only as good as PartMC's own composition realism, which the paper does not independently test.

doi: 10.5194/acp-26-10965-2026

Yee et al. 2026 Atmos Meas Tech

UCB-GLOBES: open mass-spectral library for identified and unidentified organic aerosol

~27,000 GC/EI-MS spectra of trimethylsilyl derivatives of semi-volatile organic aerosol from US and Central Amazon ambient samples plus laboratory SOA, catalogued *without regard to known chemical identity* and annotated with sample provenance and predicted properties. Reanalysis of prior ambient data gave an 11% net increase in MS-based source categorisation for the southeastern US and lifted Central Amazon matches from 136 to 375 of 1,513 analytes. **~70% of UCB-GLOBES spectra match nothing** in the library or in NIST — the honest headline is the size of the unidentified fraction, not the size of the library.

doi: 10.5194/amt-19-5027-2026

Zhao et al. 2026 Biomed Environ Sci

WHA air-LSTM: composite morbidity-driven pollution index for next-day outpatient forecasting

A five-level composite index is built *per city* from city-specific exposure–response relationships across >223.7 million hospital visits, then fed with temperature, humidity and visit history into an LSTM predicting next-day respiratory outpatient load. Beijing $R^2 = 0.963$, RMSE 53.5; removing the index degrades RMSE by 44.1%; performance holds on held-out cities. The index is fitted on morbidity and then used to predict morbidity, so the incremental-value claim is partly circular — the external validation is what carries it. Attractive template for turning a dense sensor network into an operational early-warning product.

doi: 10.3967/bes2026.062

Li et al. 2026 Front Pediatr

Interpretable ML for 90-day paediatric allergic-rhinitis exacerbation from clinical and ambient data

Development cohort $n = 2,000$ linking NHANES (2005–06, 2011–18) to EPA AQS and NOAA data; external cohort $n = 50$ from Wuhan. L2-regularised logistic regression was prespecified and beat random forest, XGBoost, LightGBM and SVM under nested 5-fold CV. Internal AUC 0.81 (0.74–0.88), external 0.71 (0.50–0.92). Baseline symptom score, same-day PM_{2.5} and total IgE dominate SHAP; **PM_{2.5} >75 vs <35 µg/m³ gives OR 5.08 (3.45–7.47).** The external cohort is 50 children on a different continent — the authors say plainly this is preliminary transportability, not external validity, and they are right.

doi: 10.3389/fped.2026.1865399

Tian et al. 2026 Atmos Meas Tech

A heterogeneity index for when lidar-retrieved CCN vertical profiles stop being trustworthy

ARM ground-based Raman lidar plus surface CCN measurements retrieve vertically resolved CCN number concentration on the assumption of vertically homogeneous composition. A k -nearest-neighbours classifier on lidar ratio and linear depolarisation ratio infers dominant aerosol type by altitude, and a derived heterogeneity index quantifies vertical complexity. Validated against HI-SCALE aircraft in situ data: **retrieval closure degrades monotonically as the index rises**, so the metric functions as a self-diagnostic flag on the retrieval. Single campaign, so the index thresholds are not yet portable.

doi: 10.5194/amt-19-5051-2026

Exposure assessment & modelling

3 records

Mushore et al. 2026 Atmos Environ X*Daily outdoor PM_{2.5} and the road-edge effect across The Gambia, Kenya and Mozambique*

Multi-temporal personal exposure measurements (2022–23) interpolated to continuous surfaces by inverse-distance weighting, validated leave-one-out at $R^2 > 0.55$, MAE $< 4 \mu\text{g}/\text{m}^3$, RMSE $< 10 \mu\text{g}/\text{m}^3$; multi-distance buffers at 10 m intervals to 1000 m from major roads. Most sites exceed WHO guidance, The Gambia highest across all land-cover classes. **The road-edge gradient reverses sign between sites** — PM_{2.5} falls with distance in Rabai and Mozambique and *risers* elsewhere, implicating wetland, fire and industrial sources that a traffic-proximity exposure proxy would misattribute entirely. IDW over sparse personal measurements is a weak spatial model; the buffer result is the transferable part.

doi: 10.1016/j.aeaoa.2026.100463

Helbich et al. 2026 Environ Int*Socioeconomic patterning of the urban external exposome depends on how the exposome is defined*

4,640 adults across Bulgaria's five largest cities, with residential NO₂, PM, traffic noise, green space, blue-space distance, impervious surface, walkability and night-time light. City-specific latent profile analysis was compared against composite indices, then both were regressed on individual- and area-level SES. High-risk profiles bundle low green space, high air pollution, noise and imperviousness *with* higher walkability. **Higher area-level SES was positively associated with high-risk profiles in most cities** — the inverse of the standard environmental-justice expectation — and composite indices only partially reproduced the latent-profile findings. The methodological warning is the finding: operationalisation choice changes the sign of the inequality conclusion.

doi: 10.1016/j.envint.2026.110419

Yadav et al. 2026 Research Square (preprint)*Airborne endotoxin on PM₁₀ in Agra: an 8-fold seasonal swing that PM mass does not track*

Full calendar year (Jan–Dec 2022) of ambient endotoxin by chromogenic LAL assay with concurrent bacterial and fungal bioaerosol, PM₁₀ and meteorology in the semi-arid Indo-Gangetic Basin. Monthly means run 0.92 ± 0.35 (June) to $7.35 \pm 1.48 \text{ EU m}^{-3}$ (December) on winter stagnation. Endotoxin correlates strongly with bacterial concentration ($r_s = 0.93$) and **not at all with PM₁₀ mass** ($r_s = 0.10$, $p = 0.75$) — mineral dust is not the carrier here. USEPA-model hazard quotients stay well below unity (max 0.0198, children, December). Preprint, single city, LAL assays are notoriously matrix-sensitive; the decoupling from mass is the durable point.

doi: 10.21203/rs.3.rs-10523351/v1

Cardiovascular & metabolic

3 records

Leiser et al. 2026 JACC Adv*Ambient PM_{2.5} and preclinical change in cardiac structure and function*

1,630 participants with echocardiography at two visits (2008–11, 2015–18), median 4.3 years apart; residential PM_{2.5} and NO₂ from spatio-temporal models as annual means preceding each visit, adjusted for baseline and time-varying covariates. Per IQR of PM_{2.5}: **global longitudinal strain +0.52% (0.20–0.83)**, LVEF $-1.05\%/ \text{year}$ (-1.73 to -0.37), **relative wall thickness +0.12 unit (0.05–0.19)**, **left atrial volume index -1.39 (-2.25 to -0.54)**. NO₂ moves the same direction, smaller. The LAVI direction sits awkwardly with a concentric-remodelling reading and the paper does not reconcile it; treat the strain and RWT results as the load-bearing ones.

doi: 10.1016/j.jacadv.2026.103104

Sewor et al. 2026 Environ Res Health*Personal household-air-pollution exposure and blood pressure in rural Rwandan children*

622 children aged 8–15 from 626 biomass-cooking households, 48 h personal PM_{2.5} and black carbon by Ultrasonic Personal Aerosol Sampler, then standardised BP measurement. Median 48 h PM_{2.5} **192.9 $\mu\text{g}/\text{m}^3$** (IQR 119.7–336.0), BC 9.85 $\mu\text{g}/\text{m}^3$. **No association with SBP or DBP** (PM_{2.5}–SBP 0.51 mmHg per 211.7 $\mu\text{g}/\text{m}^3$ IQR, 95% CI -0.46 to 1.49) and no clear effect modification by age or sex. Yet 24% of the children were hypertensive or had elevated BP against US reference percentiles. Cross-sectional, and the exposure range sits entirely above the region of the global exposure–response curve where slope is expected — a null that constrains where measurement effort is epidemiologically productive.

doi: 10.1088/2752-5309/ae8ff9

Liu et al. 2026 China CDC Wkly*Long-term ozone and cardiovascular disease: exposure windows, sex and region*

Meta-analysis of cohort studies from PubMed, Web of Science and Embase, stratified in the way most prior reviews were not. Per $10 \mu\text{g}/\text{m}^3$ ozone: CVD mortality RR 1.0105 (1.0027–1.0185), morbidity RR 1.0578 (1.0289–1.0874). The exposure-window split is the result: **warm-season ozone RR 1.0186 (1.0119–1.0253) versus an annual-average RR of 0.9971 (0.9781–1.0164)** — annual averaging destroys the signal. Female RR 1.1135 (1.0057–1.2328) versus male 1.0707 (0.9855–1.1633); Asia estimates are imprecise. Co-pollutant record; no PM estimate is reported. The averaging-window lesson transfers directly to PM network design.

doi: 10.46234/ccdcw2026.161

Reproductive & developmental

1 record

Li et al. 2026 Environ Pollut

Prenatal NO₂ and stillbirth in 3.98 million Iranian mother–baby pairs

3,982,304 pairs, 2013–2018, pregnancy-average NO₂ from satellite-based gridded estimates averaged within 10 km of the delivery hospital, analysed under both the Iranian (≥ 22 weeks) and WHO (≥ 28 weeks) stillbirth definitions. Per 10 $\mu\text{g}/\text{m}^3$: **OR 1.206 (1.192–1.220) and 1.106 (1.088–1.125)** respectively, essentially unchanged on adjustment for PM_{2.5} and ozone. The exposure–response is monotonic with *steeper slope at low concentrations*. A 10 $\mu\text{g}/\text{m}^3$ target would avert 7,686 (7,242–8,127) stillbirths, an attributable fraction of 25.8%. Exposure assigned at the delivery hospital rather than residence is a substantial misclassification channel in a country with heavy referral flows, and the effect size is large enough to warrant scepticism. Co-pollutant, not PM.

doi: 10.1016/j.envpol.2026.128908

Occupational & indoor exposure

2 records

Makanjuola 2026 Research Square (preprint)

Personal PM exposure and spirometry among 563 Southwest Nigerian street sweepers

Cross-sectional, non-smoking sweepers aged 18–60 from Ado-Ekiti, Ibadan and Ikeja; personal PM_{2.5} and PM₁₀ by handheld PMS5003 at breathing height; spirometry to ATS standard. Mean PM_{2.5} **103.55 $\pm$ 72.03 $\mu\text{g}/\text{m}^3$** and PM₁₀ 105.09 $\pm$ 73.95, roughly 20 $\times$ and 7 $\times$ the WHO 2021 annual guideline. PM_{2.5} correlates inversely with FEV₁/FVC ($r = -0.14$, $p = 0.001$) but **the association is fully attenuated in the adjusted model** ($\beta = -0.0001$, $p = 0.286$). A bare PMS5003 without RH correction at these loadings is a real measurement-error problem the paper does not address, and PM_{2.5} $\approx$ PM₁₀ is itself a red flag for the instrument. Reported for the exposure characterisation, not the health inference.

doi: 10.21203/rs.3.rs-9928829/v1

Piedra et al. 2026 Front Public Health

Informal employment versus cantonal satellite pollution as respiratory-symptom predictors

1,807 Ecuadorian workers across four sectors, ECRHS symptom instrument cross-referenced with Sentinel-5P retrievals; optimal spatial/temporal concentration percentile selected per outcome by AIC; Firth penalised logistic regression. **Informal employment dominates:** asthma aOR 12.56 (5.09–32.87), wheeze 6.78 (1.83–27.89), chronic cough 4.14 (2.26–7.74). After adjusting for cantonal altitude (itself protective, aOR 0.78, 0.66–0.93), NO₂ and SO₂ remain independent predictors of chronic phlegm and cough while ozone loses significance. Sentinel-5P carries no PM product, so the ambient arm is entirely gas-phase; cantonal assignment is coarse and the design is cross-sectional. The interesting move is using satellite retrievals as an occupational-surveillance covariate.

doi: 10.3389/fpubh.2026.1874197

Mechanistic toxicology

4 records

Silva et al. 2026 Environ Toxicol Pharmacol

PM_{2.5} organic extracts in 2D monolayers versus 3D spheroids of bronchial epithelium

16HBE cells in paired 2D and 3D culture exposed to solvent-extractable organic fractions of well-characterised ambient PM_{2.5} from three sampling periods. **Concordance between formats is endpoint-dependent, not uniform:** WST-1 metabolic activity, ROS fluorescence and extracellular metabolite exchange respond more strongly in 2D, while pro-inflammatory cytokine secretion is more evident in spheroids. Cell death is necrotic rather than apoptotic in both. The practical implication for in vitro PM toxicology is that model choice silently selects which hazard you can detect — 2D favours direct cytotoxic and oxidative readouts, 3D favours responses that need spatial organisation and exposure gradients.

doi: 10.1016/j.etap.2026.105125

Mishra et al. 2026 bioRxiv (preprint)

OMA1–DELE1 as a shared node across ultrafine PM, black carbon and nanoplastic immunotoxicity

Human PBMCs exposed to ultrafine PM, black carbon and polystyrene nanoplastics under matched conditions, profiled across NRF2, HIF-1 α , PGC-1 α , TFAM, OMA1, DELE1, mitochondrial ND1, complexes I–V, NF- κ B, TNF- α and NLRP3, with PCA, hierarchical clustering and correlation networks. All three cause oxidative stress and mitochondrial dysfunction but with distinct kinetics: **UFPM gives an acute antioxidant/adaptive response, black carbon a chronic OMA1–DELE1 integrated stress response with inflammation, nanoplastics a low-grade persistent adaptation.** The comparative design under matched dosing is what makes this worth reading; it is a preprint, expression-level only, with no functional respirometry.

doi: 10.64898/2026.07.29.741662

Wang et al. 2026 Ecotoxicol Environ Saf

Multigenerational reproductive toxicity of three vehicular PM types in C. elegans

SRM diesel PM, WHTC diesel PM and WLTP gasoline PM, differing in morphology and elemental composition, all produced dose-dependent brood-size reduction, germ-cell apoptosis, lower sperm count and morphological abnormality in parents. Mechanistically: suppressed SPE-8/SPE-9/major sperm protein/vitellogenin, activated *ced-3/-4/-9*, disrupted *set-2*, *met-2*, *spr-5*. **Effects persist to F1–F2 and recover by F3–F4 — intergenerational carryover, not stable transgenerational inheritance**, which is the distinction most papers in this space elide. Nematode model; extrapolation to mammalian germline is speculative.

doi: 10.1016/j.ecoenv.2026.120609

Sharma et al. 2026 J Toxicol Environ Health A

Dim light at night modulates the neuroimmune response to burn-pit emission inhalation

32 male Swiss Webster mice housed under dark nights or dim light at night, exposed at week 4 to filtered air or burn-pit emissions (6.9 mg/m³ mass, 4.03 $\times$ 10⁶ particles/cc) for 4 h/day over 6 days. **No behavioural differences** in open field, Y-maze or elevated plus maze. Hippocampal *Il6*, *Tnf α* , *Il1 β* and cortical *Il6* were decreased in LD-exposed mice, with reduced hippocampal *Bmal1*; lung and liver were unchanged. Small *n* per cell, a single sex, and a cytokine direction opposite to the usual inflammatory prior — reported as a circadian-interaction signal, not as an inflammation result.

doi: 10.1080/15287394.2026.2708884

Burden, policy & mitigation

5 records

Yoon et al. 2026 Med J Aust

Heat versus bushfire-smoke PM_{2.5} across four arms of the Australian health system

National ED presentations, inpatient admissions, MBS primary-care claims and PBS prescriptions, 2014–2020, for cardiovascular, respiratory and mental-health conditions, supporting Australia's first National Climate Risk Assessment. **Heat loads the front door** (ED RR 1.033 per °C for respiratory, 1.023 for mental health) while **PM_{2.5} loads admissions and prescribing** (RR 1.014 per 1 µg/m³ at 1-day lag for both cardiovascular and respiratory admissions; primary-care mental-health visits RR 1.049, 1.040–1.058). Mental-health services are sensitive to both exposures across every domain — the least-anticipated result and the one with the weakest mechanistic account. Ecological in exposure assignment.

doi: 10.5694/mja2.70261

Slavik et al. 2026 Environ Challenges

Objective versus subjective wildfire-smoke exposure and parental protective behaviour

2,086 parents in wildfire-prone western US and Canada, surveyed and linked to 3.5 years of smoke-monitoring data (2020–2023). Perceived exposure (sight or smell) tracks measured exposure but predicts behaviour far better: $b = 1.11$ (0.92–1.30) **additional protective actions for perceived exposure versus** $b = 0.23$ (0.06–0.40) **per 10 µg/m³ of measured PM_{2.5}**. The measured-exposure relationship is non-monotonic — rise, plateau, rise. Self-reported behaviour is the obvious weakness and recall is correlated with perception by construction, which inflates the contrast. Even discounted, the direction is a design constraint on any public-facing sensor network.

doi: 10.1016/j.envc.2025.101362

Wan 2026 Econ Hum Biol

Place-based ecological restoration and self-rated health: quasi-experimental evidence

Six waves of the China Family Panel Studies against the staggered prefecture-level rollout of China's national ecological restoration pilot, two-way fixed-effects DiD. **+0.1697 points of self-rated health on a five-point scale**, no differential pre-trends in event study, robust to placebo tests, Goodman–Bacon decomposition, propensity-score matching, double machine learning, IV, and adjustment for overlapping environmental policies. Mediation runs through lower PM_{2.5}, greater greenness and better visibility alongside economic and psychosocial channels. The unusually thorough robustness battery is the reason to take it seriously; the mediation analysis is correlational and the PM_{2.5} share is not separately identified.

doi: 10.1016/j.ehb.2026.101636

Zaid et al. 2026 J Mater Sci

Fibrous aerosol filtration: theory, CFD assumptions, metrology and fabrication

Review integrating single-fibre capture mechanisms and most-penetrating-particle-size theory with electrostatic charge generation, transport and stability; 3D reconstruction of filter geometry; CFD, particle tracking, deposition and clogging simulation; and electrospinning, meltblowing, bubble electrospinning and supersonic solution blowing. The section worth reading is the critique of the homogeneity and steady-state assumptions that most published simulations inherit, alongside an explicit treatment of instrumentation limits and measurement uncertainty in filtration testing. Review, no new data; the material–process–performance framing is the organising contribution.

doi: 10.1007/s10853-026-13269-8

Sidharaddi et al. 2026 Research Square (preprint)

Electrodeposited copper nodule films as electrostatic particulate collectors

Copper nodules electrodeposited onto steel; XRD confirms polycrystalline FCC, SEM a nodular microstructure, AFM a non-monotonic dependence of grain size (86.5→224.6 nm) and roughness (23.4→76.2 nm) on bath concentration before both fall again. The 0.075 M film (721 nm thick) gave **88.78% removal of dhoop smoke and an air-purifying factor of 54.12 over 405 s**, exceeding a stainless-steel discharge rod. Chamber test with a single combustion source, no reference-grade particle instrument, no size resolution and no ozone measurement — which for a corona-type collector is the omission that matters. Included as an engineering-control data point, tier C.

doi: 10.21203/rs.3.rs-10126181/v1

Other clinical endpoints

2 records

Jung et al. 2026 Ann Rheum Dis

PM_{2.5} and rheumatoid arthritis disease activity and flare

Prospective cohort at a South Korean tertiary centre, 2021–2024: 1,070 patients, 12,583 outpatient visits, monthly mean concentrations of six pollutants, generalised estimating equations adjusted for demographics, serology, medication, socioeconomic and meteorological covariates, plus a within-patient case-crossover sensitivity analysis on daily concentrations. **PM_{2.5} alone was associated with flare (adjusted OR 1.113, 95% CI 1.017–1.218)** and with higher DAS28-CRP, CDAI, and tender and swollen joint counts; the association was stronger in women and non-smokers, and the case-crossover arm reproduced it for cumulative exposure beyond two weeks. Single centre, monitor-assigned exposure; the two-design agreement is what makes it convincing.

doi: 10.1016/j.ard.2026.06.018

Lees 2026 Research Square (preprint)

PM_{2.5} component mixtures and inpatient hospitalisation in chronic kidney disease

2,511 *All of Us* participants with ICD-coded CKD, 19,656 hospitalisations over 22,448 person-years (2010–2019); NASA SEDAC estimates of organic and elemental carbon, ammonium, nitrate and sulfate linked at 3-digit ZIP; weighted quantile sum regression inside a negative binomial model. Median PM_{2.5} was only 4.4 µg/m³. **IRR 1.31 (0.99–1.73, $p = 0.06$) per quartile of the WQS index — formally null, with 93% of the mixture weight on nitrate**, 4% each on ammonium and elemental carbon, and zero on organic carbon and sulfate. WQS weights are unstable at this sample size and 3-digit ZIP assignment is very coarse. The nitrate concentration of weight is the hypothesis worth carrying, not the point estimate.

doi: 10.21203/rs.3.rs-10438586/v1

Corpus register

First author	Focus	Journal	Design	Metric	Region
Cardiovascular & metabolic					
Leiser et al.	Ambient PM _{2.5} and preclinical re-modelling of cardiac structure and function	<i>JACC Adv</i>	Observational - cohort	PM _{2.5} and NO ₂ (spatio-temporal model, annual mean at residence)	North America
Liu et al. (CCDC)	Long-term ozone and cardiovascular disease: warm-season vs annual-average exposure windows	<i>China CDC Wkly</i>	Review / synthesis	Ozone (co-pollutant; no PM estimate reported)	Global / multi-region
Sewor et al.	Null: personal HAP exposure and blood pressure in rural Rwandan children at 193 µg/m ³ median PM _{2.5}	<i>Environ Res Health</i>	Observational - cross-sectional	PM _{2.5} and black carbon (48 h personal, Ultrasonic Personal Aerosol Sampler)	Sub-Saharan Africa
Reproductive & developmental					
Li et al. (Environ Pollut)	Prenatal NO ₂ and stillbirth in 3.98 million Iranian mother-baby pairs	<i>Environ Pollut</i>	Observational - cohort	NO ₂ (satellite-based gridded, 10 km buffer); PM _{2.5} and O ₃ as co-pollutants	Middle East & N. Africa
Mechanistic toxicology					
Mishra et al.	OMA1-DELE1 as the shared node across ultrafine PM, black carbon and nanoplastic immunotoxicity	<i>bioRxiv (preprint)</i>	Experimental / toxicology	Ultrafine PM, black carbon, polystyrene nanoplastics (matched conditions)	South Asia
Sharma et al.	Dim light at night reshapes the neuroimmune response to burn-pit emission inhalation	<i>J Toxicol Environ Health A</i>	Experimental / toxicology	Burn-pit emissions, 6.9 mg/m ³ mass and 4.03e6 particles/cc	North America
Silva et al.	2D monolayers and 3D spheroids disagree by endpoint, not by magnitude, for PM _{2.5} organic extracts	<i>Environ Toxicol Pharmacol</i>	Experimental / toxicology	Solvent-extractable organic fraction of characterised ambient PM _{2.5}	Latin America
Wang et al.	Vehicular PM reproductive toxicity carries to F2 but not F3 in <i>C. elegans</i>	<i>Ecotoxicol Environ Saf</i>	Experimental / toxicology	SRM diesel PM, WHTC diesel PM, WLTP gasoline PM	China
Occupational & indoor					
Makanjuola	Street sweepers at 20x the WHO PM _{2.5} guideline, with spirometry that does not follow	<i>Research Square (preprint)</i>	Observational - cross-sectional	Personal PM _{2.5} and PM ₁₀ (PMS5003 handheld, breathing height)	Sub-Saharan Africa
Piedra et al.	Informal employment dwarfs cantonal satellite pollution as a respiratory-symptom predictor	<i>Front Public Health</i>	Observational - cross-sectional	Sentinel-5P NO ₂ , SO ₂ , O ₃ (cantonal percentiles; no PM product)	Latin America
Sensing, forecasting & instrumentation					
Lei et al.	Eight-year speciated PM ₁ record in Dublin: heating emissions vs trans-boundary transport	<i>Atmos Chem Phys</i>	Measurement campaign	PM ₁ (chemically speciated, ACSM + BC)	Europe
Li et al. (Front Pediatr)	Interpretable ML for 90-day paediatric allergic-rhinitis exacerbation from clinical + ambient exposure	<i>Front Pediatr</i>	Modelling / inventory	PM _{2.5} , PM ₁₀ , O ₃ , NO ₂ (same-day and 0-7 d lag)	North America
Liu et al. (ACP)	From kappa to chi: PartMC evaluation of hygroscopicity-based mixing-state retrieval	<i>Atmos Chem Phys</i>	Modelling / inventory	HTDMA kappa distributions; mixing-state index chi (50-250 nm)	Global / multi-region
Somnugpong	When seasonal refinement helps (and does not help) Stineman imputation of gappy air-quality series	<i>Environ Monit Assess</i>	Modelling / inventory	PM _{2.5} (UCI Air Quality; Beijing PM _{2.5})	China
Tian et al.	When lidar-based CCN profiles fail: a heterogeneity index for vertically layered aerosol	<i>Atmos Meas Tech</i>	Measurement campaign	CCN number concentration; lidar ratio and depolarisation ratio	North America

First author	Focus	Journal	Design	Metric	Region
Yee et al.	UCB-GLOBES: open-access GC/EI-MS spectral library for identified and unidentified organic aerosol	<i>Atmos Meas Tech</i>	Measurement campaign	Semi-volatile organic aerosol (TMS derivatives, GC/EI-MS)	Global / multi-region
Zhao et al.	WHA air-LSTM: composite morbidity-driven pollution index for next-day respiratory outpatient forecasting	<i>Biomed Environ Sci</i>	Modelling / inventory	Composite air pollution index (WHA air), city-specific ER curves	China
Exposure assessment & modelling					
Helbich et al.	Socioeconomic patterning of the urban external exposome depends on how you define it	<i>Environ Int</i>	Observational - cross-sectional	PM and NO2 within an 8-exposure external exposome	Europe
Mushore et al.	Daily outdoor PM _{2.5} exposure and the road-edge effect across three sub-Saharan campaigns	<i>Atmos Environ X</i>	Measurement campaign	PM _{2.5} (personal, interpolated to surfaces)	Sub-Saharan Africa
Yadav et al.	Airborne endotoxin on PM ₁₀ in Agra: an 8-fold seasonal swing that PM mass does not track	<i>Research Square (preprint)</i>	Measurement campaign	PM ₁₀ mass; endotoxin (LAL); bacterial and fungal bioaerosol	South Asia
Burden, policy & mitigation					
Sidharaddi et al.	Electrodeposited copper nodule films as electrostatic PM collectors	<i>Research Square (preprint)</i>	Experimental / toxicology	Dhoop-smoke particulate (chamber, unspecified)	South Asia
Slavik et al.	Perceived smoke beats measured smoke as a predictor of parental protective action	<i>Environ Chall</i>	Observational - cross-sectional	PM _{2.5} on smoke days (monitoring data) vs self-reported sight/smell	North America
Wan	Ecological restoration as a population health intervention, with PM _{2.5} as one mediator	<i>Econ Hum Biol</i>	Ecological	PM _{2.5} concentration (mediator alongside greenness and visibility)	China
Yoon et al.	Heat vs bushfire-smoke PM _{2.5} across four arms of the Australian health system	<i>Med J Aust</i>	Ecological	PM _{2.5} (bushfire smoke) and temperature, short-term lags	Oceania
Zaid et al.	Fibrous aerosol filtration: where CFD assumptions and measurement uncertainty diverge from practice	<i>J Mater Sci</i>	Review / synthesis	Ultrafine particles; most penetrating particle size; pressure drop	Global / multi-region
Other clinical endpoints					
Jung et al.	PM _{2.5} and rheumatoid arthritis flare: 12,583 visits with a case-crossover sensitivity arm	<i>Ann Rheum Dis</i>	Observational - cohort	PM _{2.5} , PM ₁₀ and four gases (monthly means; daily for case-crossover)	East Asia (ex-China)
Lees	Null with a nitrate signal: PM _{2.5} component mixtures and hospitalisation in CKD	<i>Research Square (preprint)</i>	Observational - cohort	PM _{2.5} components OC, EC, NH ₄ , NO ₃ , SO ₄ (NASA SEDAC, 3-digit ZIP)	North America

Provenance and method

Entry window **6 August 2026**, contiguous with the 5 August issue; no gap. Most shipped records carry a 6 August PubMed entry date, Europe PMC creation date or Crossref created date; five PubMed records carry a 4–5 August entry date and are **explicitly flagged backfill** — PubMed re-indexed them into today's window and `seen.json` confirms they were never shipped. They must be excluded from any weekly trend line. **Sources.** PubMed E-utilities and the PubMed connector on two separate axes (health, instrumentation) windowed on [EDAT]: the connector returned 20 health and 4 sensing hits, the local harvester 14 and 4, and **the two disagree in both directions** — ten records were connector-only and three harvester-only, so both legs are merged before screening. Union 25 records, 18 carried, 7 rejected (health-literacy instruments, e-cigarette nanoparticles, cigarette-smoke miRNA, marine particulate organic carbon, dredged-sludge stabilisation, a COPD protease review with no PM exposure, and one duplicate already shipped on 5 August). Europe PMC on both `FIRST_PDATE` and `CREATION_DATE` — 7 preprints, 5 carried, 1 rejected as off-topic (GPU lattice-Boltzmann radiative transfer), 1 dropped as already shipped. Crossref scoped by journal ISSN across eight aerosol and atmospheric titles PubMed does not index — 11 hits, 4 carried, 5 rejected out of scope, 2 held for want of an abstract. Consensus on three sensing axes (calibration and drift, satellite-surface fusion, humidity correction and edge inference) — **all nine hits resolved to DOIs created between January and June 2026**, i.e. entirely outside this window, and were rejected rather than backfilled; Ginsburg et al.'s global PurpleAir calibration is flagged in the run log as the priority pickup if a backfill slot opens. OpenAlex returned HTTP 429 on top of the existing paid block. arXiv returned 60 records of which 2 postdate 25 July, neither in scope after the recency screen. Scholar Gateway was queried for the held Prajapati hygroscopicity record and returned adjacent literature only — its corpus is current to May 2026. **Screening.** Deduplication keys PMID → lowercased DOI → title signature, checked against `seen.json` before summarising. Consensus titles were resolved to DOIs through Crossref before screening. **Nine records are held, not rejected:** DOIs verify in Crossref but no abstract is retrievable from Crossref, Europe PMC, Semantic Scholar or Consensus. Seven are carried over from 4–5 August and were re-tried on this run without success (a third attempt for four of them); two are new Elsevier deposits. Held records stay out of `seen.json` so a later issue can carry them. Nothing was summarised from a title alone. **Labels.** Subtopic, design, particle-metric and geography assignments are single-label judgements from title, abstract, keywords and MeSH terms, not a controlled vocabulary. The six effect estimates are transcribed verbatim; two are co-pollutant (NO₂, O₃) rather than PM and are marked as such in the figure caption and the register. All DOI links resolve to the publisher of record and passed the automated DOI gate (0 fail, 7 advisory title-overlap warnings, all from editorial focus lines) before this issue compiled. Abstracts remain the copyright of their publishers.